Micro Reentry Detection :)

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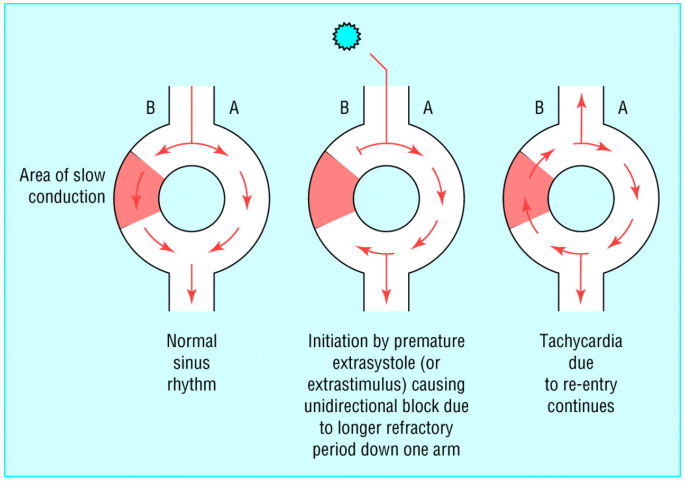
Overview

Basic Reentry

- A reentry circuit is a circulating excitation wave in a closed pathway of excitable tissue.
- **Structure:** Loop (anatomical or functional) with cell-to-cell conduction.
- **Sustaining Condition and Key Dynamics:**
  - Wave returns after tissue has recovered excitability
  - Requires sufficient path length relative to conduction delay and refractory period
  - Favored by tissue heterogeneity (e.g., areas of slow conduction)



Micro Reentry vs Macro Reentry

- **Macro-reentry:** [?]
  - Large-scale circuit around anatomical structures (e.g., valves, scars, **pulmonary veins**)
  - Path length is long relative to wavelength ($\lambda = v \cdot \text{ERP}$)
  - Typically mappable with clinical techniques (e.g., ECG, catheter mapping)
- **Micro-reentry:** [?]
  - Small, localized circuits driven by tissue heterogeneity
  - Path length comparable to wavelength (near minimum for sustained reentry)
  - Often below clinical resolution; inferred via modeling or high-density mapping

Overview of Micro Reentry

- **Definition:** Micro reentry circuits are small-scale reentrant pathways in excitable media (e.g., cardiac tissue) that sustain self-propagating electrical activity. [**?**]
- **Mechanism:**
  - Requires a closed conduction loop
  - Unidirectional block in part of the pathway
  - Conduction delay allowing tissue ahead to recover excitability
- **Characteristic Scale:**
  - Typically on the order of centimeters or smaller
  - Often below spatial resolution of clinical imaging
- **Physiological Context:**
  - Commonly observed in cardiac arrhythmias (e.g., atrial fibrillation, ventricular tachycardia)
  - Associated with heterogeneous tissue properties (fibrosis, ischemia)

Why Study Reentry?

- **Primary mechanism of arrhythmias:**
  - Reentry underlies many tachyarrhythmias (e.g., atrial fibrillation, ventricular tachycardia) [**?**, **?**]
- **Guides therapeutic intervention:**
  - Identification of reentrant pathways enables targeted ablation strategies [**?**]
- **Links structure to function:**
  - Tissue heterogeneity (fibrosis, infarct) promotes reentry formation and stability [**?**]
- **Enables predictive modeling:**
  - Computational models of reentry support patient-specific diagnosis and treatment planning [**?**]

Micro Reentry After Ablation

- Microreentrant atrial tachycardias can occur after AF ablation, especially at gaps in ablation lines [**?**].
- These circuits are very small—sometimes just a few millimeters—and often missed with traditional mapping [**?**].
- High-resolution mapping enables detection and targeted ablation of these circuits.

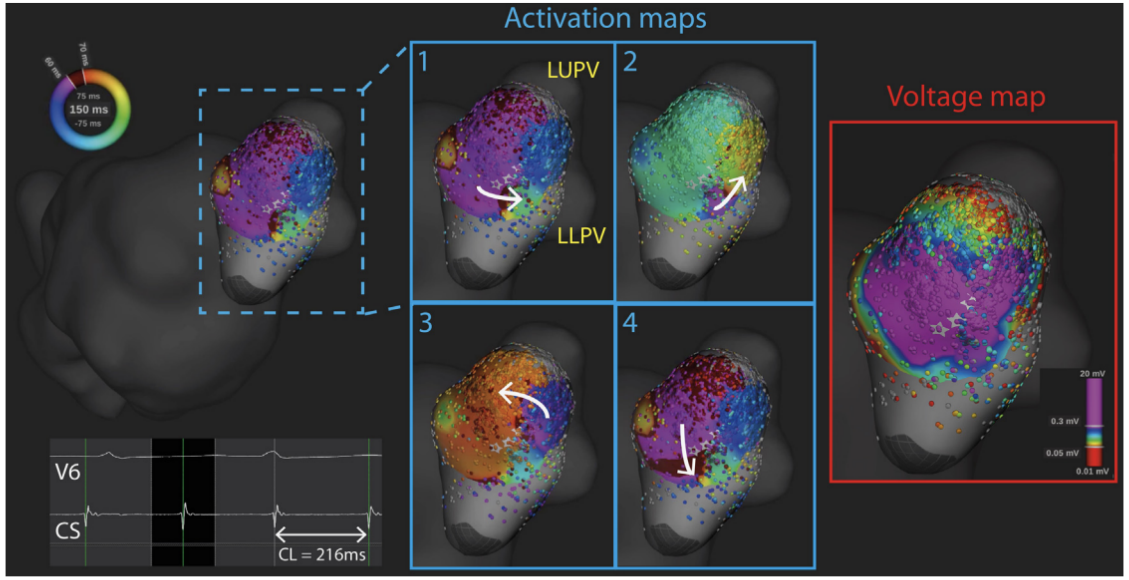


Figure 2 Microreentrant atrial tachycardia anterior to left-sided pulmonary veins. During mapping, the patient developed an atrial tachycardia (cycle length [CL] 220 ms). A high-density Rhythmia map was performed around the left pulmonary veins, which demonstrated a microreentrant circuit anterior to the left pulmonary veins, near the site of the gap in the wide area circumferential ablation line in Figure 1. The 4 panels are sequential activation maps demonstrating the microreentrant circuit. The corresponding bipolar voltage map during atrial tachycardia is also shown. LLPV = left lower pulmonary vein; LUPV = left upper pulmonary vein.

Size of Microreentrant Circuits

- Microreentrant circuits can be extremely small—sometimes just a few millimeters, far below conventional mapping resolution and detectable only with high-density systems like Rhythmia-Orion [**?**, **?**].
- The likelihood and location of microreentry depend on tissue structure, wall thickness, and fibrosis.

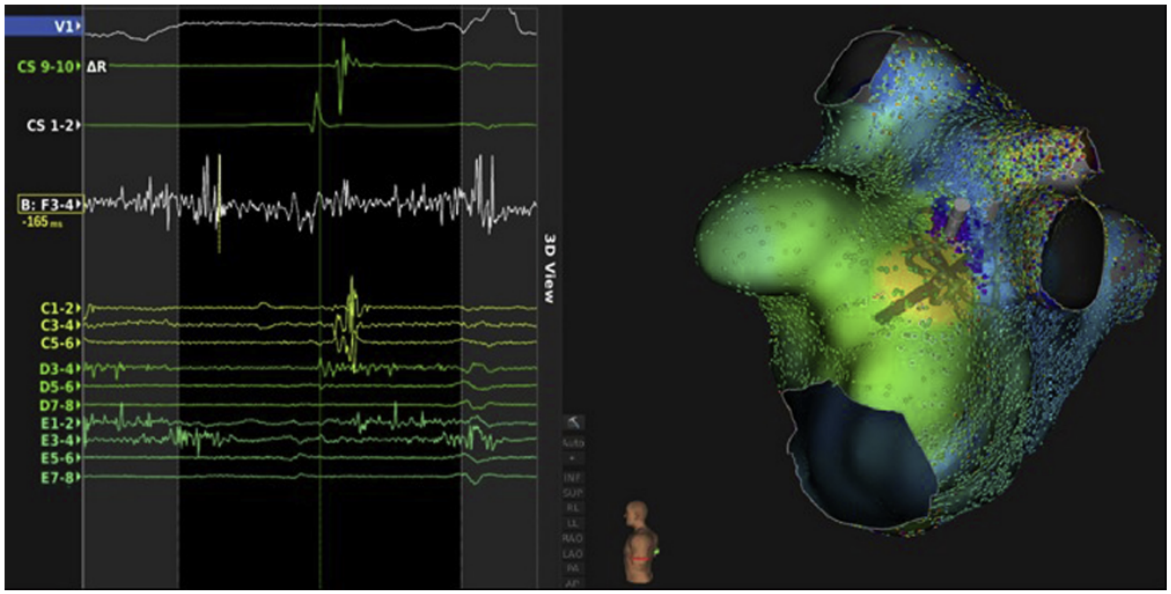


Figure 2 The Orion (Boston Scientific, Marlborough, MA) bipolar pair F3–4 (white in the picture) detects a continuous highly fragmented signal that spans the entire cycle length, showing that the microreentrant tachycardia is confined to a region size of the order of millimeters.

What we are doing

Our Goal

- **Objective:**
  - Develop a computational framework to automatically detect reentrant activity in excitable networks
- **Input Parameters:**
  - Network topology (muscular connectivity graph)
  - Conduction delays along edges
  - Node refractory periods
- **Core Goal:**
  - Identify conditions under which circulating excitation (reentry) emerges from initial stimulation
- **Technical Focus:**
  - Detect closed-loop propagation satisfying timing constraints (return after recovery)
  - Characterize stability and persistence of reentrant pathways

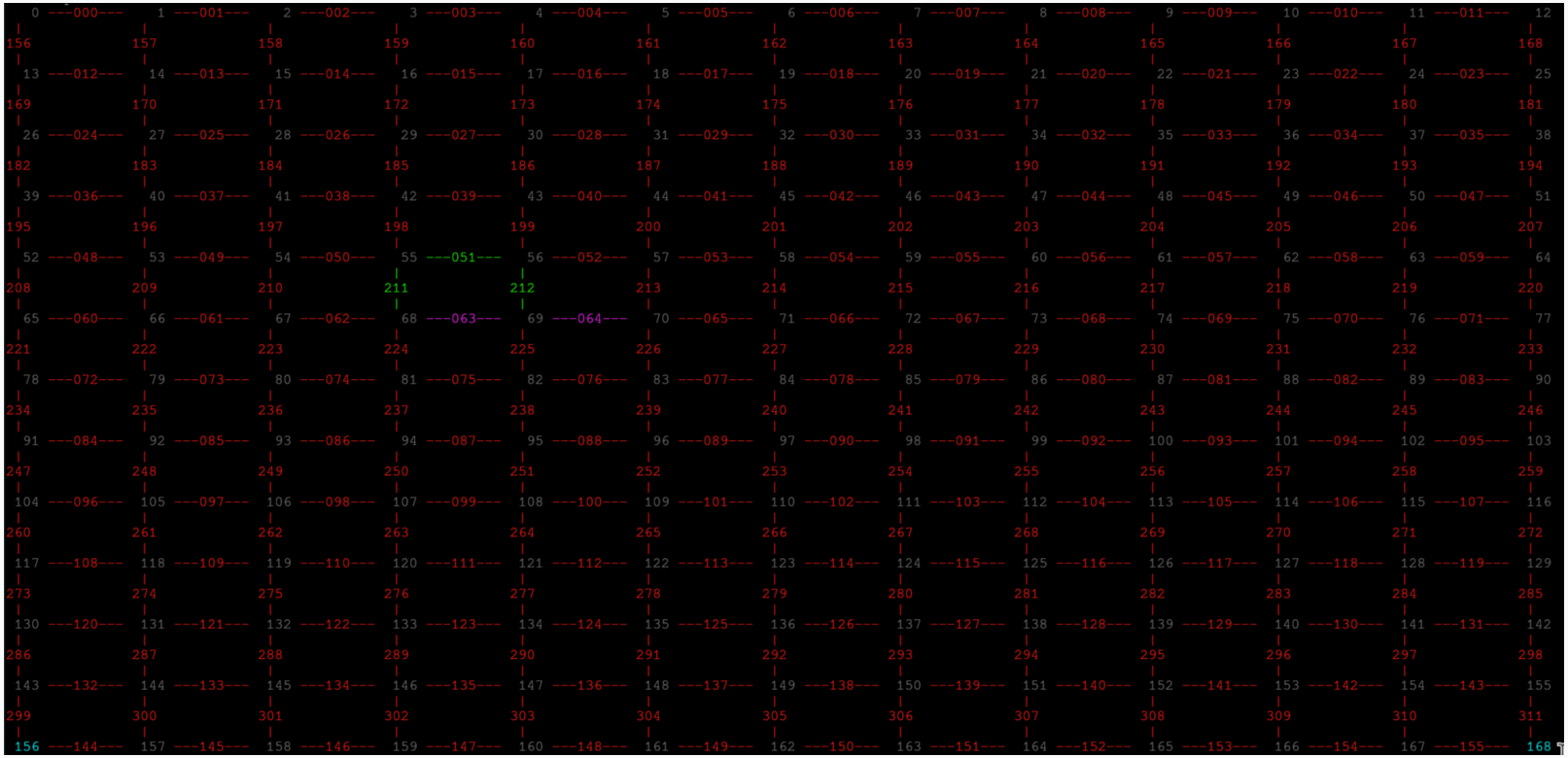
How does this goal fit with the other projects we have?

- **Integration with Existing Work:**
  - Builds on top of the main simulator to enable a more simplistic way to search for reentrany circuit dynamics. The long term goal is to use this tool to determine what the pattern is for reentrant circuits and to make some connections to scar tissue or fibrosis patterns.
- **Shared Parameters, Extended Search Dynamics**
  - Utilizes the same network properties such as RP and CT while increasing the generality of the simulation.
  - Spatial dimentionality has also been removed to consider solely the connections between nodes and their properties.
  - Other methods of examing circuits are also enabled such as monte carlo simulations.

Simple mesh program

- **Simulator Overview:**
  - Discrete-time excitable network simulator modeling electrical propagation through cardiac-like tissue
  - Designed to enable automatic search for network properties- currently focused on reentry detection
- **Initialization & Control:**
  - Configurable initial stimulation patterns and network parameters
  - Supports parameter sweeps across conduction and refractory properties
- **Outputs:**
  - Setup to record network properties based on initial configuration parameters that result in a reentrant circuit
  - Any output record can be viewed using just a terminal

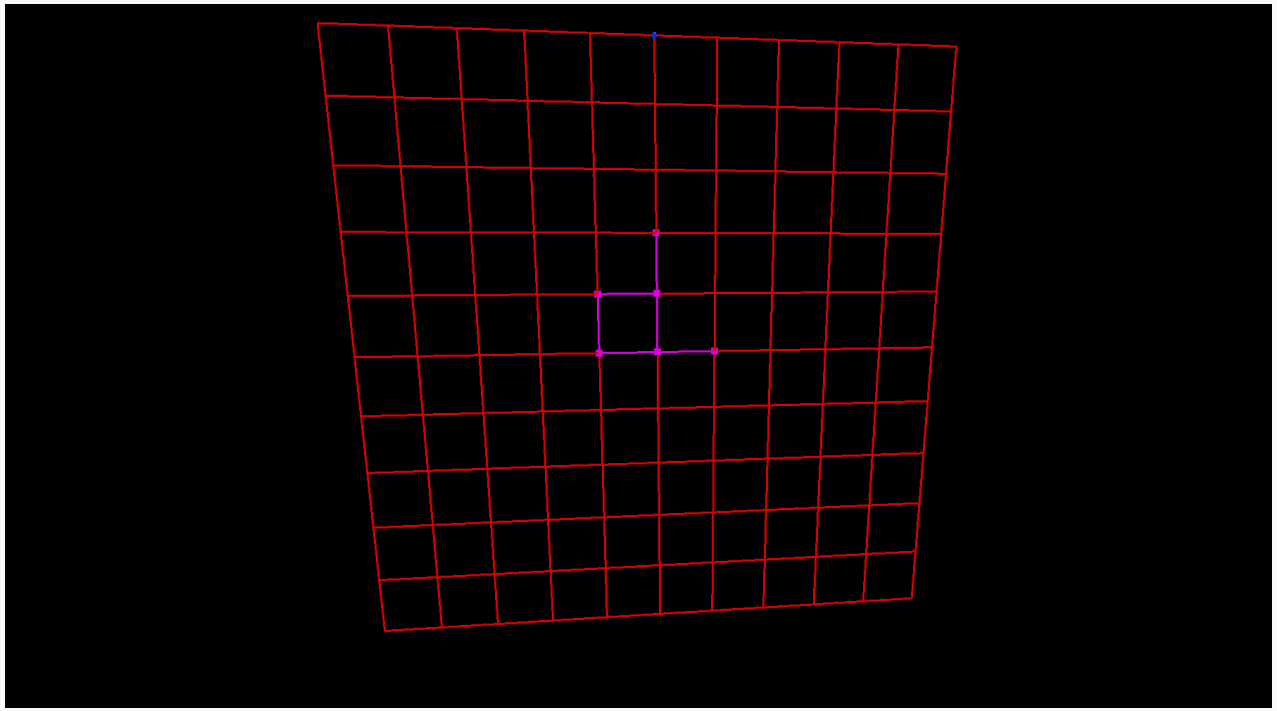
Basic Mesh Program Setup



- We created a simple version of our model in Python that is a 12X12 grid. The muscles and nodes have ID's that we can change the Conduction Time(CT) and the Refractory Period(RP). This allows us to test simulations much quick to get base values.

Results

Basic Setup



- We tested several different shapes by hand to see what the smallest shape that would produce a micro reentry circuit would be. We found it to be this box with 2 appendages. We used the basic values of $RP = 0.04$ and $CV = 0.3$

Current Results

- An initial trial run of 1000 Monte Carlo simulations with RP ranges of (0.01, 0.1) and a CT range of (3.0,4.0) in our initial simple box yielded a 64.9% reentry rate.
- These results were transferable into the main simulator, although not every successful run was checked due to how long it takes to setup in the main sim
- Future Monte Carlo sims will include both exploring different ranges of values and different sets of muscles to modify

Transferring setups from the mini-sim to full model

- **Model Integration:**
  - Results are stored in JSON format and include the origal paremeters for the monte carlo sim, as well as each trial result
  - Trial results include the specific parameters for each modified muscle and whether that trial resulted in a reentrant circuit
- **Interface to Simulator:**
  - Currently models need to be recreated by hand. Long term it should not be difficult to write a script to generate the files needed to run in the main simulator.
  - In addition, the mini-sim used CT rather than CV and Length due to not using a spatial model, so this needs to be accounted for when transferring models.
- **Scalability Considerations:**
  - Currently designed primarily for relatively small networks. However, due to not including things like contraction, it does not require a significant amount of computational resources

Current Limitations

- Currently it is difficult to efficiently move results between simlulators. It should not be too difficult to fix this in the future.
- This simulator is primarily designed for running small scale simulations. It is doubtful that it would perform well for larger meshes.

Future plans for the project

- Perform more monte carlo sims across a wider range of sub-networks (That is, differnt sets of muscles that get modified)
- Automate the process of moving sims to the main model
- Use the simulation results to determine a general rule or pattern for micro circuits in this type of model and use that pattern to draw conculsions about micro reentry circutis in the real heart

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